PHYS 435 · Exam II · FRONT: ELECTRIC (sections 1–6)

$k = \frac{1}{4\pi\epsilon_0}$; boxed = result; every card: setup → solve → result

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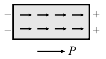
1 · E, D, P: FIELDS IN DIELECTRICS

GENERAL SETUP



E = total field from ALL charge: $\nabla\cdot\mathbf{E} = \frac{\rho_f+\rho_b}{\epsilon_0}$, $\nabla\times\mathbf{E}=0$, $\mathbf{E}=-\nabla V$

P = dipole moment per volume (C/m²)

Bound charge: $\sigma_b = \mathbf{P}\cdot\hat{n}$ ($\hat{n}$ out of the material), $\rho_b = -\nabla\cdot\mathbf{P}$

$\mathbf{D} \equiv \epsilon_0\mathbf{E}+\mathbf{P}$ (C/m²): $\nabla\cdot\mathbf{D} = \rho_f$, $\oint \mathbf{D}\cdot d\mathbf{a} = Q_{f,enc}$

$\nabla\times\mathbf{D} = \nabla\times\mathbf{P}$ (NOT zero in general)

USEFUL FACTS

Linear: $\mathbf{P} = \epsilon_0\chi_e\mathbf{E}$ and $\mathbf{D} = \epsilon\mathbf{E}$

$\epsilon = \epsilon_0(1+\chi_e) = \epsilon_0\epsilon_r$; $\epsilon_r = 1+\chi_e$; $\chi_e = \epsilon_r - 1$

Vacuum: $\chi_e = 0$, $\epsilon_r = 1$, $\epsilon = \epsilon_0$

$\mathbf{P} = \mathbf{D}-\epsilon_0\mathbf{E} = \frac{\chi_e}{\epsilon_r}\mathbf{D} = (1-\frac{1}{\epsilon_r})\mathbf{D}$

$\mathbf{E} = \mathbf{D}/\epsilon$ (linear); $\mathbf{E} = (\mathbf{D}-\mathbf{P})/\epsilon_0$ (frozen P: no ε!)

Homogeneous linear: $\rho_b = -\frac{\chi_e}{1+\chi_e}\rho_f$, so no free charge inside ⇒ no ρ_b

Next to a conductor: $\sigma_f + \sigma_b = \sigma_f/\epsilon_r$

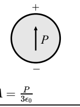
$\boxed{\text{No free charge + symmetry} \Rightarrow \mathbf{D}=0 \Rightarrow \mathbf{E} = -\mathbf{P}/\epsilon_0 \text{ inside, 0 outside}}$

Total bound charge is always 0 (use it as a check)

$\boxed{\text{Recipe: D (Gauss, free charge)} \to \text{E = D/}\varepsilon \to \text{P} \to \sigma_b, \rho_b \to \text{V from } \infty \text{ inward}}$

SPECIFIC RESULTS (setup → solve → boxed result)

1.1 Uniformly polarized sphere



Setup: uniform P ⇒ $\rho_b = 0$, $\sigma_b = \mathbf{P}\cdot\hat{r} = P\cos\theta$: a shell with $\sigma = P\cos\theta$

Solve: $V_{in} = Ar\cos\theta$, $V_{out} = \frac{B\cos\theta}{r^2}$

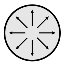
V continuous: $AR = \frac{B}{R^2}$

jump: $\partial_r V_{out} - \partial_r V_{in} = -\frac{\sigma}{\epsilon_0} \Rightarrow -\frac{2B}{R^3} - A = -\frac{P}{\epsilon_0} \Rightarrow A = \frac{P}{3\epsilon_0}$

$\boxed{\mathbf{E}_{in} = -\frac{\mathbf{P}}{3\epsilon_0} \text{ (uniform)}}$ $\boxed{\text{outside: dipole } \mathbf{p} = \frac{4}{3}\pi R^3\mathbf{P}}$

$V_{in} = \frac{P}{3\epsilon_0}r\cos\theta$, $V_{out} = \frac{PR^3\cos\theta}{3\epsilon_0 r^2}$

1.2 Ball with radial P = k r HW 4.10



Setup: $\sigma_b = \mathbf{P}\cdot\hat{r}|_R = kR$; $\rho_b = -\frac{1}{r^2}\frac{d}{dr}(r^2\cdot kr) = -3k$

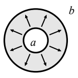
Solve: no free charge, spherical symmetry ⇒ D = 0 ⇒ $\mathbf{E} = -\mathbf{P}/\epsilon_0$

check: Gauss with $Q_{enc} = -3k\cdot\frac{4}{3}\pi r^3$ gives the same

$\boxed{\rho_b = -3k, \sigma_b = kR, \mathbf{E}_{in} = -k\mathbf{r}/\epsilon_0, E_{out} = 0}$

Any $P = kr^n\hat{r}$: $\rho_b = -(n+2)kr^{n-1}$, $\sigma_b = kR^n$

1.3 Shell a<r<b with P = (k/r) r̂ Gr 4.15

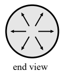


Setup: $\rho_b = -\frac{1}{r^2}\frac{d}{dr}(r^2\frac{k}{r}) = -\frac{k}{r^2}$; outer face $\hat{n} = \hat{r}$; inner face $\hat{n} = -\hat{r}$

Solve: D = 0 everywhere (no free charge, symmetry) ⇒ $\mathbf{E} = -\mathbf{P}/\epsilon_0$ in the shell

$\boxed{\sigma_b(b) = \frac{k}{b}, \sigma_b(a) = -\frac{k}{a}, E = -\frac{k}{\epsilon_0 r}\hat{r} \text{ in shell, 0 elsewhere}}$

1.4 Long cylinder with P = k s ŝ



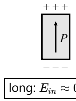
Setup: $\rho_b = -\frac{1}{s}\frac{d}{ds}(s\cdot ks) = -2k$; $\sigma_b = P(a) = ka$

Solve: D = 0 by cylindrical symmetry ⇒ $\mathbf{E} = -\mathbf{P}/\epsilon_0$

$\boxed{\rho_b = -2k, \sigma_b = ka, E_{in} = -\frac{ks}{\epsilon_0}, \text{out } 0}$

Any $ks^n\hat{s}$: $\rho_b = -(n+1)ks^{n-1}$; uniform P ⊥ axis: $\mathbf{E}_{in} = -\frac{\mathbf{P}}{2\epsilon_0}$

1.5 Uniform P in a cylinder (bar electret) HW 4.11



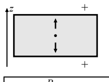
Setup: $\rho_b = 0$; $\sigma_b = +P$ on top cap, $-P$ on bottom cap, 0 on the side

Solve: field of two charged disks ±P: long rod ⇒ two point charges $\pm P\pi a^2$; short disk ⇒ capacitor

$\boxed{\text{long: } E_{in} \approx 0\text{; short: } E_{in} \approx -P/\epsilon_0}$

Shape factor: $\mathbf{E}_{in} = -N\mathbf{P}/\epsilon_0$; N = 1 (slab ⊥), ⅓ (sphere), ½ (cyl ⊥), 0 (needle ∥)

1.6 Slab with P = P₀(z/a) ẑ, |z|<a

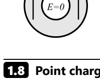


Setup: $\rho_b = -\frac{dP_z}{dz} = -\frac{P_0}{a}$; top: $P_z(a) = P_0$; bottom: $\hat{n} = -\hat{z}$, $P_z(-a) = -P_0 \Rightarrow +P_0$

Solve: planar symmetry, no free charge ⇒ D = 0 ⇒ $E = -P/\epsilon_0$

$\boxed{\rho_b = -\frac{P_0}{a}, \sigma_b = +P_0 \text{ on both faces, } E_z = -\frac{P_0 z}{a\epsilon_0} \text{ inside, 0 outside}}$

1.7 Cavities and shells of uniform P



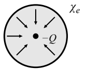
Setup: cavity = full object with +P plus a cavity-sized ball with −P

Solve: add the inside fields: $-\frac{\mathbf{P}}{3\epsilon_0} + \frac{\mathbf{P}}{3\epsilon_0}$

$\boxed{\text{uniformly polarized shell: E = 0 in the cavity}}$

cavity in a P-slab (P ⊥): $-\frac{\mathbf{P}}{\epsilon_0} + \frac{\mathbf{P}}{3\epsilon_0} = -\frac{2\mathbf{P}}{3\epsilon_0}$

1.8 Point charge at the center of a dielectric ball



(charge −Q at center, susceptibility χ_e, radius R, no other free charge)

Setup: Gaussian sphere of radius r < R: $Q_{f,enc} = -Q$

Solve: $D\cdot 4\pi r^2 = -Q \to \mathbf{E} = \mathbf{D}/\epsilon \to \mathbf{P} = \frac{\chi_e}{1+\chi_e}\mathbf{D} \to \sigma_b = \mathbf{P}\cdot\hat{r}|_R \to \rho_b = -\nabla\cdot\mathbf{P}$

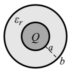
$\boxed{\mathbf{D} = -\frac{Q}{4\pi r^2}\hat{r}}$ $\boxed{\mathbf{E} = -\frac{Q}{4\pi\epsilon_0(1+\chi_e)r^2}\hat{r}}$

$\boxed{\mathbf{P} = -\frac{\chi_e Q}{4\pi(1+\chi_e)r^2}\hat{r}}$ $\boxed{\sigma_b(R) = -\frac{\chi_e Q}{4\pi(1+\chi_e)R^2}}$

$\boxed{\rho_b = 0 \text{ for r > 0, plus a point charge } +\frac{\chi_e Q}{1+\chi_e} \text{ at the origin}}$

(since $\nabla\cdot\frac{\hat{r}}{r^2} = 4\pi\delta^3(\mathbf{r})$); check: total bound = 0; outside $E = -\frac{Q}{4\pi\epsilon_0 r^2}$

1.9 Conductor (Q, radius a) inside a dielectric shell to b HW 4.26



Setup: Gaussian sphere r > a: $Q_{f,enc} = Q$

Solve: $D = \frac{Q}{4\pi r^2}$; $E = \frac{D}{\epsilon}$ in the shell, $\frac{D}{\epsilon_0}$ outside; $V(a) = \int_a^\infty E\,dr$ split at b

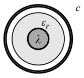
$\sigma_b = \mathbf{P}\cdot\hat{n}$ with $\hat{n} = -\hat{r}$ at a, $+\hat{r}$ at b

$\boxed{V(a) = kQ[\frac{1}{b} + \frac{1}{\epsilon_r}(\frac{1}{a} - \frac{1}{b})]}$

$\boxed{\sigma_b(a) = -\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi a^2}, \sigma_b(b) = +\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi b^2}}$

Two shells: at the interface $\sigma_b = \frac{Q}{4\pi b^2}(\frac{1}{\epsilon_{r2}} - \frac{1}{\epsilon_{r1}})$

1.10 Coated wire / coax with dielectric

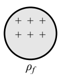


Setup: Gaussian cylinder radius s, length L: $Q_{f,enc} = \lambda L$

Solve: $D = \frac{\lambda}{2\pi s} \to E = \frac{D}{\epsilon} \to V = \int E\,ds \to \sigma_b = \mathbf{P}\cdot\hat{n}$

$\boxed{E = \frac{\lambda}{2\pi\epsilon s}; \sigma_b(a) = -\frac{\chi_e}{\epsilon_r}\frac{\lambda}{2\pi a}; \Delta V = \frac{\lambda}{2\pi\epsilon}\ln\frac{b}{a}}$

1.11 Ball of dielectric holding free charge ρ_f



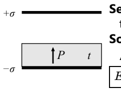
Setup: Gaussian sphere r: $Q_{f,enc} = \rho_f\frac{4}{3}\pi r^3$

Solve: $D = \frac{\rho_f r}{3} \to E = \frac{\rho_f r}{3\epsilon} \to P = \frac{\chi_e}{\epsilon_r}D \to \sigma_b = P(R)$

$\boxed{\rho_b = -\frac{\chi_e}{\epsilon_r}\rho_f, \sigma_b = \frac{\chi_e}{\epsilon_r}\frac{\rho_f R}{3}}$

Free charge only on the surface instead ⇒ inside D = E = P = 0

1.12 Frozen P slab inside a capacitor (±σ)



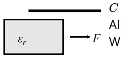
Setup: pillbox from inside the + plate: D = σ in the slab AND in the gap (planar)

Solve: $E = (D-P)/\epsilon_0$ in slab, D/ϵ_0 in gap; $\Delta V = \sum E\times$thickness

$\boxed{E_{slab} = \frac{\sigma-P}{\epsilon_0}, E_{gap} = \frac{\sigma}{\epsilon_0}, \Delta V = \frac{\sigma d - Pt}{\epsilon_0}}$

2 · CAPACITORS & ENERGY

GENERAL SETUP



$C = Q/\Delta V$; $W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$

All charge: $W = \frac{\epsilon_0}{2}\int E^2 d\tau = \frac{1}{2}\int \rho V\,d\tau$

With dielectric (work on free charge): $W = \frac{1}{2}\int \mathbf{D}\cdot\mathbf{E}\,d\tau = \frac{1}{2}\int\frac{D^2}{\epsilon}d\tau$

USEFUL FACTS

Stacked layers = series (same D): $\frac{1}{C} = \sum\frac{1}{C_i}$

Side by side = parallel (same E): $C = \sum C_i$

Fill with ϵ_r: fixed Q ⇒ W ÷ ϵ_r; fixed V ⇒ W × ϵ_r

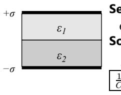
$\boxed{\text{Force: } F = \frac{1}{2}V^2\frac{dC}{dx}\text{, toward higher C}}$

Conductor surface pressure: $\frac{\sigma^2}{2\epsilon_0}$; energy outside a sphere: $\frac{Q^2}{8\pi\epsilon_0 R}$

Infinite line/cylinder: energy per length diverges outside

SPECIFIC RESULTS (setup → solve → boxed result)

2.1 Parallel plates, stacked slabs HW 4.18



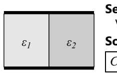
Setup: pillbox from inside the + plate ⇒ $D = \sigma$ in every layer

Solve: $E_i = \frac{\sigma}{\epsilon_i}$; $V = \sum E_i d_i$; $C = \frac{\sigma A}{V}$; $P_i = \sigma(1-\frac{1}{\epsilon_{ri}})$; interface $\sigma_b = P_1 - P_2$

$\boxed{\frac{1}{C} = \sum\frac{d_i}{\epsilon_i A}\text{; half-filled: } C = \frac{2\epsilon_r}{1+\epsilon_r}C_0}$

4.18 (ϵ_r 2 and 1.5, each thickness a): $V = \frac{7\sigma a}{6\epsilon_0}$, σ_b(interface) $= \frac{\sigma}{6}$

2.2 Parallel plates, side-by-side slabs Disc 4.19

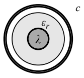


Setup: both halves span the same plates ⇒ same V ⇒ $E = V/d$ in each

Solve: $\sigma_{f,i} = \epsilon_i E$; $Q = \sum\sigma_{f,i}A_i$; $C = Q/V$

$\boxed{C = \sum\frac{\epsilon_i A_i}{d}\text{; half-filled: } C = \frac{1+\epsilon_r}{2}C_0}$

2.3 Coax HW 4.21



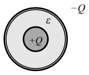
Setup: $D = \frac{\lambda}{2\pi s}$ in every layer

Solve: $V = \sum\frac{\lambda}{2\pi\epsilon_i}\ln\frac{b_i}{a_i}$; $\frac{C}{L} = \frac{\lambda}{V}$

$\boxed{\frac{C}{L} = \frac{2\pi\epsilon_0}{\sum\ln(b_i/a_i)/\epsilon_{ri}}}$

Half-filled by angle (E shared): $\frac{C}{L} = \frac{\pi\epsilon_0(1+\epsilon_r)}{\ln(b/a)}$

2.4 Spherical capacitor



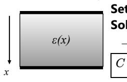
Setup: $D = \frac{Q}{4\pi r^2}$

Solve: $V = \sum\frac{Q}{4\pi\epsilon_i}(\frac{1}{a_i} - \frac{1}{b_i})$; $C = Q/V$

$\boxed{C = \frac{4\pi\epsilon ab}{b-a}\text{; isolated sphere } 4\pi\epsilon_0 R}$

$\epsilon = \epsilon_0\frac{b}{r}$: $C = \frac{4\pi\epsilon_0 b}{\ln(b/a)}$

2.5 Permittivity that varies



Setup: D = σ still (free charge only)

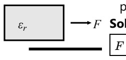
Solve: $E = \frac{\sigma}{\epsilon(x)}$; $V = \sigma\int_0^d\frac{dx}{\epsilon}$; $P = D - \epsilon_0 E$; $\rho_b = -\frac{dP}{dx}$ (not 0: ε varies)

$\boxed{C = \frac{A}{\int_0^d dx/\epsilon}}$

$\epsilon = \epsilon_0(1+\frac{x}{d})$: $C = \frac{\epsilon_0 A}{d\ln 2}$

linear $\epsilon_1 \to \epsilon_2$: $C = \frac{A(\epsilon_2-\epsilon_1)}{d\ln(\epsilon_2/\epsilon_1)}$

2.6 Dielectric pulled into a capacitor

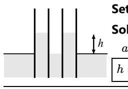


Setup: inserted length x: two capacitors in parallel, $C = \frac{\epsilon_0 w}{d}(\ell + \chi_e x)$

Solve: $F = \frac{1}{2}V^2\frac{dC}{dx}$

$\boxed{F = \frac{\epsilon_0\chi_e wV^2}{2d}\text{, pulled in}}$

2.7 Oil rising in a coax Gr 4.28

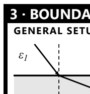


Setup: $C(h) = \frac{2\pi\epsilon_0(L+\chi_e h)}{\ln(b/a)}$

Solve: $F = \frac{1}{2}V^2\frac{dC}{dh} = \frac{\pi\epsilon_0\chi_e V^2}{\ln(b/a)}$ = weight $\rho g\pi(b^2 - a^2)h$

$\boxed{h = \frac{\epsilon_0\chi_e V^2}{\rho g(b^2-a^2)\ln(b/a)}}$

2.8 Energy integrals



Setup: pick a route: $\frac{1}{2}QV$ (fastest), or $\frac{1}{2}\int\frac{D^2}{\epsilon}d\tau$ with $d\tau = 4\pi r^2 dr$

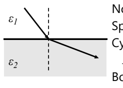
Solve: one region at a time; outside a sphere always $\frac{Q^2}{8\pi\epsilon_0 R}$

$\boxed{\text{shell } \frac{Q^2}{8\pi\epsilon_0 R}\text{; ball } \frac{3Q^2}{20\pi\epsilon_0 R}\text{; ball of } \epsilon_r\text{: } \frac{Q^2}{8\pi\epsilon_0 R}(1+\frac{1}{5\epsilon_r})}$

4.26: $\frac{Q^2}{8\pi\epsilon_0\epsilon_r}(\frac{1}{a} + \frac{\chi_e}{b})$; coax per length $\frac{\lambda^2}{4\pi\epsilon}\ln\frac{b}{a}$

3 · BOUNDARY CONDITIONS & LAPLACE PROBLEMS

GENERAL SETUP



No charge in a region ⇒ $\nabla^2 V = 0$

Sphere: $V = \sum(A_l r^l + \frac{B_l}{r^{l+1}})P_l(\cos\theta)$

Cylinder: $A_0 + B_0\ln s + \sum(A_n s^n + B_n s^{-n})(C_n\cos n\phi + D_n\sin n\phi)$

Box: $e^{\pm kx}\sin ky$, sinh, cosh

BCs: $D_1^\perp - D_2^\perp = \sigma_f$; $\mathbf{E}^\parallel$ continuous; V continuous

No free σ: $\epsilon_1\partial_n V_1 = \epsilon_2\partial_n V_2$

Charged sheet in vacuum: $\partial_n V_{out} - \partial_n V_{in} = -\sigma/\epsilon_0$

USEFUL FACTS

Center finite ⇒ drop $r^{-(l+1)}$, s^{-n}, $\ln s$

Far away ⇒ drop r^l, except $-E_0 r\cos\theta$ for a uniform field

Uniform field ⇒ only l = 1 (n = 1) terms appear

$P_0 = 1$, $P_1 = \cos\theta$, $P_2 = \frac{3\cos^2\theta - 1}{2}$, $P_3 = \frac{5\cos^3\theta - 3\cos\theta}{2}$

$\cos^2\theta = \frac{1}{3} + \frac{2}{3}P_2$; $c_l = \frac{2l+1}{2}\int_{-1}^1 f(x)P_l(x)\,dx$

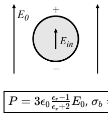
Field lines bend away from the normal entering high ε: $\frac{\tan\theta_1}{\tan\theta_2} = \frac{\epsilon_1}{\epsilon_2}$

Uniqueness: anything that meets every BC is THE answer

$\boxed{\text{Recipe: regions} \to \text{general V} \to \text{kill terms} \to \text{number BCs} \to \text{match each } P_l \to \text{check limits}}$

SPECIFIC RESULTS (setup → solve → boxed result)

3.1 Dielectric sphere in a uniform field E₀



Setup: $V_{in} = Ar\cos\theta$, $V_{out} = -E_0 r\cos\theta + \frac{B\cos\theta}{r^2}$

BCs at R: (i) V continuous (ii) $\epsilon_r\partial_r V_{in} = \partial_r V_{out}$

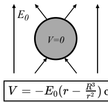
Solve: (i) $AR = -E_0 R + \frac{B}{R^2}$ (ii) $\epsilon_r A = -E_0 - \frac{2B}{R^3}$; eliminate B

$\boxed{A = -\frac{3E_0}{\epsilon_r+2}, E_{in} = \frac{3E_0}{\epsilon_r+2}\text{ uniform, } B = \frac{\epsilon_r-1}{\epsilon_r+2}E_0R^3}$

$\boxed{P = 3\epsilon_0\frac{\epsilon_r-1}{\epsilon_r+2}E_0, \sigma_b = P\cos\theta}$

Check: $\epsilon_r = 1$ ⇒ vacuum; $\epsilon_r \to \infty$ ⇒ conductor

3.2 Conducting sphere in a uniform field E₀



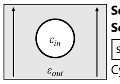
Setup: $V = -E_0 r\cos\theta + \frac{B\cos\theta}{r^2}$; BC: V(R) = 0

Solve: $B = E_0R^3$; $\sigma = -\epsilon_0\partial_r V|_R$

$\boxed{V = -E_0(r - \frac{R^3}{r^2})\cos\theta, \sigma = 3\epsilon_0 E_0\cos\theta, p = 4\pi\epsilon_0 R^3 E_0}$

Net charge Q: add $\frac{kQ}{r}$. Cylinder ⊥: $V = -E_0(s - \frac{R^2}{s})\cos\phi$, $\sigma = 2\epsilon_0 E_0\cos\phi$

3.3 Bubble, or any ε_in inside ε_out



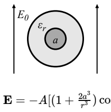
Setup: same as 3.1 with ϵ_{in}, ϵ_{out} in BC (ii)

Solve: $A = -\frac{3\epsilon_{out}E_0}{\epsilon_{in}+2\epsilon_{out}}$

$\boxed{\text{sphere } E_{in} = \frac{3\epsilon_{out}}{\epsilon_{in}+2\epsilon_{out}}E_0\text{; air cavity } \frac{3\epsilon_r}{1+2\epsilon_r}E_0}$

Cylinder ⊥: $\frac{2\epsilon_{out}}{\epsilon_{in}+\epsilon_{out}}E_0$; dielectric cylinder $\frac{2E_0}{\epsilon_r+1}$

3.4 Coated conductor in a uniform field HW 4.24



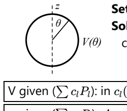
Setup: shell: $V = (Ar + \frac{C}{r^2})\cos\theta$ with V(a) = 0 ⇒ $C = -Aa^3$; outside $-E_0 r\cos\theta + \frac{B\cos\theta}{r^2}$

Solve: at b: V continuous and $\epsilon_r\partial_r V_{shell} = \partial_r V_{out}$; eliminate B

$\boxed{A = \frac{-3E_0}{(\epsilon_r+2)+2(\epsilon_r-1)a^3/b^3}}$

$\mathbf{E} = -A[(1+\frac{2a^3}{r^3})\cos\theta\hat{r} - (1-\frac{a^3}{r^3})\sin\theta\hat{\theta}]$

3.5 Spherical shell with V(θ) or σ(θ)



Setup: inside $A_l r^l P_l$; outside $\frac{B_l}{r^{l+1}}P_l$

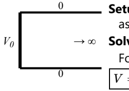
Solve: expand the given function in P_l; V continuous at R gives $B_l = A_l R^{2l+1}$

$\boxed{\text{V given } (\sum c_l P_l)\text{: in } c_l(\frac{r}{R})^l\text{, out } c_l(\frac{R}{r})^{l+1}, \sigma = \frac{\epsilon_0}{R}\sum(2l+1)c_l P_l}$

$\boxed{\sigma \text{ given } (\sum\sigma_l P_l)\text{: } A_l = \frac{\sigma_l}{(2l+1)\epsilon_0 R^{l-1}}}$

$\sigma = k\cos\theta$: $E_{in} = \frac{k}{3\epsilon_0}$, pointing down

3.6 Slot / box (Cartesian)



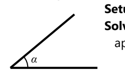
Setup: V = X(x)Y(y); BCs: V = 0 at y = 0, a; V → 0 as x → ∞; V = V_0 at x = 0

Solve: zero walls ⇒ $\sin\frac{n\pi y}{a}$; decay ⇒ $e^{-n\pi x/a}$; Fourier: $C_n = \frac{2}{a}\int_0^a V_0\sin\frac{n\pi y}{a}dy$

$\boxed{V = \frac{4V_0}{\pi}\sum_{n\,odd}\frac{1}{n}e^{-n\pi x/a}\sin\frac{n\pi y}{a}}$

closed at x = b: $\frac{\sinh(n\pi(b-x)/a)}{\sinh(n\pi b/a)}$; symmetric: cosh

3.7 Wedge / pipe (polar)



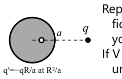
Setup: $V = R(r)\Theta(\theta)$; walls at θ = 0, α grounded

Solve: $\Theta = \sin\frac{n\pi\theta}{\alpha}$, $R = Ar^k + Br^{-k}$ with $k = \frac{n\pi}{\alpha}$; apply the arcs

$\boxed{\text{zero arc at a: } (\frac{r}{a})^k - (\frac{a}{r})^k\text{; one wall at } V_0\text{: } V = \frac{V_0\theta}{\alpha}}$

4 · IMAGES

GENERAL SETUP



Replace the conductor (or dielectric) by fictitious charges placed OUTSIDE the region you solve in

If V matches on every boundary and at ∞, uniqueness ⇒ correct in that region

USEFUL FACTS

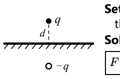
Induced charge: $\sigma = -\epsilon_0\frac{\partial V}{\partial n}$; total induced = sum of the images

Force on q: from the images only

Energy to bring q in: ½ × q × (V of the images)

SPECIFIC RESULTS (setup → solve → boxed result)

4.1 Charge above a grounded plane



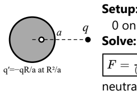
Setup: image −q at the mirror point (V = 0 on the plane)

Solve: $V = kq(\frac{1}{r_1} - \frac{1}{r_2})$; $\sigma = -\epsilon_0\partial_z V|_{z=0}$

$\boxed{F = \frac{kq^2}{4d^2}\text{ (pull); } \sigma = -\frac{qd}{2\pi(s^2+d^2)^{3/2}}; W = -\frac{kq^2}{4d}}$

Plane held at V_0: add V_0

4.2 Charge near a grounded sphere



Setup: image $q' = -\frac{qR}{a}$ at $\frac{R^2}{a}$ from the center (V = 0 on the sphere)

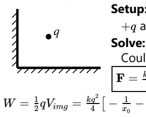
Solve: $F = \frac{kqq'}{(a-R^2/a)^2}$

$\boxed{F = \frac{kq^2Ra}{(a^2-R^2)^2}\text{ (pull)}}$

neutral sphere: add $+\frac{qR}{a}$ at center; sphere at V_0: add $4\pi\epsilon_0 RV_0$ at center

q inside a grounded shell at d: image $-\frac{qR}{d}$ at $\frac{R^2}{d}$

4.3 Charge in a 90° corner



Setup: images: $-q$ at $(-x_0, y_0)$, $-q$ at $(x_0, -y_0)$, $+q$ at $(-x_0, -y_0)$

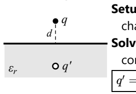
Solve: force on q = vector sum of the three Coulomb forces

$\boxed{\mathbf{F} = \frac{kq^2}{4}\left[-\frac{\hat{x}}{x_0^2} - \frac{\hat{y}}{y_0^2} + \frac{x_0\hat{x}+y_0\hat{y}}{(x_0^2+y_0^2)^{3/2}}\right]}$

$W = \frac{1}{2}qV_{img} = \frac{kq^2}{4}\left[-\frac{1}{x_0} - \frac{1}{y_0} + \frac{1}{\sqrt{x_0^2+y_0^2}}\right]$

Wedge of angle π/n: 2n − 1 images (only these angles work)

4.4 Charge above a dielectric half-space



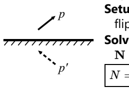
Setup: above: q + image q' at −d; below: one charge q'' at +d

Solve: at z = 0: V continuous ⇒ $q + q' = q''$; $\epsilon\partial_z V$ continuous ⇒ $q - q' = \epsilon_r q''$

$\boxed{q' = -\frac{\epsilon_r-1}{\epsilon_r+1}q, q'' = \frac{2q}{\epsilon_r+1}}$

$\boxed{F = \frac{kqq'}{(2d)^2}; \sigma_b = -\frac{1}{2\pi}\frac{\chi_e}{\chi_e+2}\frac{qd}{(s^2+d^2)^{3/2}}}$

4.5 Dipole above a grounded plane HW 4.6

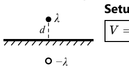


Setup: image each charge: component ∥ to plane flips, ⊥ component kept

Solve: field of the image at p (distance 2z), then $\mathbf{N} = \mathbf{p}\times\mathbf{E}_{img}$

$\boxed{N = \frac{kp^2\sin\theta\cos\theta}{(2z)^3}\text{; rests ⊥ to the plane}}$

4.6 Line charge above a plane

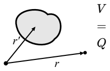


Setup: image $-\lambda$ (dielectric: $\times\frac{\epsilon_r-1}{\epsilon_r+1}$)

$\boxed{V = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{s_-}{s_+}\text{; force/length } \frac{\lambda^2}{4\pi\epsilon_0 d}}$

5 · MULTIPOLES & DIPOLES

GENERAL SETUP



$$V = k\sum_n\frac{1}{r^{n+1}}\int r'^n P_n(\cos\alpha)\rho\,d\tau'$$
$$= k[\frac{Q}{r} + \frac{\mathbf{p}\cdot\hat{r}}{r^2} + \frac{1}{r^3}\sum q_i r_i^2 P_2(\cos\alpha_i) + \ldots]$$
$$Q = \int\rho\,d\tau;\ \mathbf{p} = \int\mathbf{r}'\rho\,d\tau' = \sum q_i\mathbf{r}_i$$

USEFUL FACTS

Far away the first nonzero moment wins: $V \propto r^{-(n+1)}$, $E \propto r^{-(n+2)}$

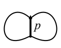
Shift origin by **a**: $\mathbf{p}' = \mathbf{p} - Q\mathbf{a}$ (Q = 0 ⇒ p is origin-independent)

$\boxed{\text{Dipole in a field: } \mathbf{N} = \mathbf{p}\times\mathbf{E}, U = -\mathbf{p}\cdot\mathbf{E}, \mathbf{F} = \nabla(\mathbf{p}\cdot\mathbf{E})}$

Uniform E: F = 0; an aligned dipole is pulled toward stronger E

SPECIFIC RESULTS (setup → solve → boxed result)

5.1 Field of a dipole



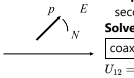
Setup: $V = \frac{kp\cos\theta}{r^2}$

Solve: $E_r = -\partial_r V$, $E_\theta = -\frac{1}{r}\partial_\theta V$

$\boxed{\mathbf{E} = \frac{kp}{r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta}) = \frac{k}{r^3}[3(\mathbf{p}\cdot\hat{r})\hat{r} - \mathbf{p}]}$

on axis $\frac{2kp}{r^3}$; on equator $-\frac{kp}{r^3}$

5.2 Forces between dipoles and charges HW 4.7



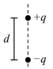
Setup: $U = -\mathbf{p}_2\cdot\mathbf{E}_1$ (field of the first at the second)

Solve: $\mathbf{F} = -\nabla U$; on an axis $F = p_2\frac{dE_1}{dz}$

$\boxed{\text{coaxial dipoles: } F = \frac{3p_1p_2}{2\pi\epsilon_0 r^4}\text{ (attract)}}$

$U_{12} = \frac{k}{r^3}[\mathbf{p}_1\cdot\mathbf{p}_2 - 3(\mathbf{p}_1\cdot\hat{r})(\mathbf{p}_2\cdot\hat{r})]$; aligned dipole near q: $F = \frac{2kqp}{r^3}$

5.3 Computing dipole moments



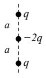
Setup: $\mathbf{p} = \sum q_i\mathbf{r}_i$ or $\int\mathbf{r}\rho\,d\tau$; use the symmetry axis

Solve: $p_z = \int z\,dq$ with $z = r\cos\theta$, $dq = \sigma R^2 d\Omega$ or $\rho r^2 dr\,d\Omega$

$\boxed{\pm\text{q apart d: } qd\text{; shell } k\cos\theta\text{: } \frac{4}{3}\pi kR^3\text{; ball } k\cos\theta\text{: } \frac{\pi kR^4}{3}}$

hemishell σ: $\pi\sigma R^3$; ring $\lambda_0\cos\phi$: $\pi R^2\lambda_0$; rod $\lambda_0 z/L$ on [−L, L]: $\frac{2}{3}\lambda_0 L^2$

5.4 Linear quadrupole (q, −2q, q)



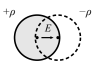
Setup: Q = 0 and p = 0 ⇒ the quadrupole term leads

Solve: $\frac{k}{r^3}\sum q_i r_i^2 P_2(\cos\alpha_i)$

$\boxed{V = \frac{kqa^2(3\cos^2\theta - 1)}{r^3}}$

force in a nonuniform E: add $q_i\mathbf{E}(\mathbf{r}_i)$

5.5 Overlapping balls ±ρ



Setup: inside a uniform ball centered at **c**: $\mathbf{E} = \frac{\rho}{3\epsilon_0}(\mathbf{r} - \mathbf{c})$

Solve: add the two; the **r** terms cancel

$\boxed{\text{overlap: } \mathbf{E} = -\frac{\rho\mathbf{d}}{3\epsilon_0}\text{ uniform (d from − center to + center)}}$

cylinders: $-\frac{\rho\mathbf{d}}{2\epsilon_0}$; d → 0 with ρd = P gives the polarized sphere

6 · MATH

Div/curl one-liners

$F(r)\hat{r}$: div $= \frac{1}{r^2}(r^2F)'$

$F(s)\hat{s}$: div $= \frac{1}{s}(sF)'$

$F(s)\hat{\phi}$: curl $= \frac{1}{s}(sF)'\,\hat{z}$

$F(s)\hat{z}$: curl $= -F'\,\hat{\phi}$

$F(z)\hat{x}$: curl $= F'\,\hat{y}$

Cylindrical

$\nabla\cdot\mathbf{A} = \frac{1}{s}\partial_s(sA_s) + \frac{1}{s}\partial_\phi A_\phi + \partial_z A_z$

$\nabla\times\mathbf{A} = [\frac{1}{s}\partial_\phi A_z - \partial_z A_\phi]\hat{s} + [\partial_z A_s - \partial_s A_z]\hat{\phi} + \frac{1}{s}[\partial_s(sA_\phi) - \partial_\phi A_s]\hat{z}$

$\nabla^2 f = \frac{1}{s}\partial_s(s\partial_s f) + \frac{1}{s^2}\partial_\phi^2 f + \partial_z^2 f$

Spherical

$\nabla\cdot\mathbf{A} = \frac{1}{r^2}\partial_r(r^2A_r) + \frac{\partial_\theta(\sin\theta A_\theta)}{r\sin\theta} + \frac{\partial_\phi A_\phi}{r\sin\theta}$

$\nabla^2 f = \frac{1}{r^2}\partial_r(r^2\partial_r f) + \frac{\partial_\theta(\sin\theta\,\partial_\theta f)}{r^2\sin\theta}$

$\nabla f = \partial_r f\,\hat{r} + \frac{1}{r}\partial_\theta f\,\hat{\theta} + \frac{\partial_\phi f}{r\sin\theta}\hat{\phi}$

$\nabla\times\mathbf{A} = \frac{\partial_\theta(\sin\theta A_\phi) - \partial_\phi A_\theta}{r\sin\theta}\hat{r} + \frac{1}{r}\left[\frac{\partial_\phi A_r}{\sin\theta} - \partial_r(rA_\phi)\right]\hat{\theta} + \frac{1}{r}\left[\partial_r(rA_\theta) - \partial_\theta A_r\right]\hat{\phi}$

Identities: $\nabla\times\nabla f = 0$; $\nabla\cdot(\nabla\times\mathbf{A}) = 0$; $\oint\mathbf{v}\cdot d\mathbf{a} = \int\nabla\cdot\mathbf{v}\,d\tau$

Unit vectors

$\hat{z} = \cos\theta\hat{r} - \sin\theta\hat{\theta}$; at the equator $\hat{\theta} = -\hat{z}$

$\hat{s} = \cos\phi\hat{x} + \sin\phi\hat{y}$; $\hat{\phi} = -\sin\phi\hat{x} + \cos\phi\hat{y}$

$\hat{s}\times\hat{\phi} = \hat{z}$, $\hat{\phi}\times\hat{z} = \hat{s}$, $\hat{z}\times\hat{s} = \hat{\phi}$, $\hat{r}\times\hat{\theta} = \hat{\phi}$

Integrals

$\int\frac{dz}{(z^2+s^2)^{3/2}} = \frac{z}{s^2\sqrt{z^2+s^2}}$

$\int\frac{z\,dz}{(z^2+s^2)^{3/2}} = -\frac{1}{\sqrt{z^2+s^2}}$

$\int\frac{dz}{\sqrt{z^2+s^2}} = \ln(z + \sqrt{z^2+s^2})$

$\int_0^\pi\sin^3\theta\,d\theta = \frac{4}{3}$; $\int_0^\pi\cos^2\theta\sin\theta\,d\theta = \frac{2}{3}$

$\int_0^\infty x^n e^{-x/a}dx = n!\,a^{n+1}$

$\int_{-1}^1 P_l^2 dx = \frac{2}{2l+1}$; $\int_0^1 P_l\,dx = 1, \frac{1}{2}, 0, -\frac{1}{8}$

Units

P, D, σ: C/m²; E: V/m; p: C·m; $\epsilon_0 = 8.85\times10^{-12}$; $k = 8.99\times10^9$

PHYS 435 · Exam II · BACK: MAGNETIC (sections 7–12) + EXAM I BASICS (13)

$\mu_0 = 4\pi \times 10^{-7}$; RHR: thumb along I, fingers curl with B

CONTENTS: find your setup, go to its number

7 CURRENTS, FORCES, BIOT–SAVART
7: setup, facts
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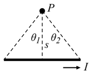
7.2 Arcs + straight leads (hairpin)
7.3 Circular loop, on its axis
7.4 Square / polygon / rectangle, at the center
7.5 Finite solenoid, on axis
7.6 Forces between currents
8 AMPÈRE & SUPERPOSITION
8: setup, facts
8.1 Solid wire, uniform J
8.2 Wire with J ∝ sⁿ; thick tube
8.3 Coax
8.4 Infinite current sheet K
8.5 Moving capacitor (two opposite sheets)
8.6 Thick slab with J x̂, |z|<a
8.7 Solenoid / toroid
8.8 Cylinder with an off-axis hole
8.9 Overlapping wires ±J; slab with a hole
9 VECTOR POTENTIAL A
9: setup, facts
9.1 Wire with J = k sⁿ ẑ, A(0) = 0
9.2 Uniform wire, everywhere
9.3 Solenoid / uniform B
9.4 Spinning shell, dipole, magnetized sphere
9.5 Sheet and slab
9.6 Given A: find B, then J
10 MAGNETIC DIPOLES & SPINNING CHARGE
10: setup, facts
10.1 Two dipoles
10.2 Spinning disk (σ, ω)
10.3 Spinning shell (σ, ω)
10.4 Spinning solid ball (ρ, ω)
10.5 Spinning cylinders
11 B, H, M: MAGNETIC MATTER
11: setup, facts
11.1 Uniformly magnetized sphere
11.2 Magnetized rod / disk
11.3 Cylinder with M along φ̂
11.4 Cylinder with M = k s ẑ
11.5 Linear slab carrying free J₀ x̂, |z|<d/2
11.6 Coax / toroid / solenoid with χₘ
11.7 Linear sphere / slab in uniform B₀
12 MAGNETIC BOUNDARY CONDITIONS & E ↔ B
12: setup, facts
13 EXAM I BASICS: COULOMB, GAUSS, CONDUCTORS
13: setup, facts
13.1 Point charge / ball
13.2 Line / cylinder
13.3 Sheet / slab
13.4 Ring / disk / segment, on axis
13.5 Overlaps and cavities

7 · CURRENTS, FORCES, BIOT–SAVART

GENERAL SETUP



$I = \lambda v$ (A); $\mathbf{K} = \sigma \mathbf{v}$ (A/m); $\mathbf{J} = \rho \mathbf{v}$ (A/m²)

$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$; on a wire $\mathbf{F} = \int I\, d\mathbf{l} \times \mathbf{B}$

$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{I\, d\mathbf{l} \times (\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|^3}$ (or $\mathbf{K}\, da$, $\mathbf{J}\, d\tau$)

Steady currents: $\nabla \cdot \mathbf{J} = 0$

USEFUL FACTS

RHR: thumb along I, fingers curl with B

A straight piece whose LINE passes through P gives 0

Arc of angle φ, radius R, at its center: $\frac{\mu_0 I \phi}{4\pi R}$

Semi-infinite wire ending beside P (distance s): $\frac{\mu_0 I}{4\pi s}$

Spinning: disk $K = \sigma\omega s$; shell $K = \sigma\omega R \sin\theta$; ball $J = \rho\omega r \sin\theta$; ring $I = \frac{Q\omega}{2\pi}$

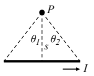
Uniform B: closed loop F = 0, torque $\mathbf{m} \times \mathbf{B}$; curved wire = straight chord

Cyclotron: $r = \frac{mv}{qB}$, $\omega = \frac{qB}{m}$; B does no work

Recipe: chop into pieces → drop pieces on P's line → table → RHR → add

SPECIFIC RESULTS (setup → solve → boxed result)

7.1 Straight segment



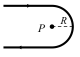
Setup: $dB = \frac{\mu_0 I}{4\pi} \frac{dz \cos\theta}{z^2+s^2}$ with $z = s\tan\theta$

Solve: $B = \frac{\mu_0 I}{4\pi s} \int_{\theta_1}^{\theta_2} \cos\theta\, d\theta$

$\boxed{B = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1) \text{ (θ from the ⊥ foot); infinite } \frac{\mu_0 I}{2\pi s}}$

wire ∥ z at (x_0, y_0): $\mathbf{B} = \frac{\mu_0 I}{2\pi} \frac{(-(y-y_0), x-x_0, 0)}{(x-x_0)^2+(y-y_0)^2}$

7.2 Arcs + straight leads (hairpin)



Setup: arc: every $d\mathbf{l} \perp \mathbf{R}$, all at distance R; leads: segment formula

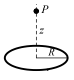
Solve: $B_{arc} = \frac{\mu_0 I}{4\pi R^2} R\phi$; add leads with RHR signs

$\boxed{\text{loop center } \frac{\mu_0 I}{2R}\text{; hairpin } \frac{\mu_0 I}{4R} + \frac{\mu_0 I}{2\pi R}}$

two arcs (a, b) + radial: $\frac{\mu_0 I \phi}{4\pi}(\frac{1}{a} - \frac{1}{b})$

7.3 Circular loop, on its axis

HW 5.37



Setup: $d\mathbf{B} \perp$ the line to $d\mathbf{l}$; only the z part survives: factor $\frac{R}{\sqrt{R^2+z^2}}$

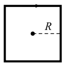
Solve: $B = \frac{\mu_0 I}{4\pi} \frac{2\pi R}{R^2+z^2} \frac{R}{\sqrt{R^2+z^2}}$

$\boxed{B = \frac{\mu_0 I R^2}{2(R^2+z^2)^{3/2}}}$

far: $\frac{\mu_0 I R^2}{2z^3}$ (dipole $m = I\pi R^2$); Helmholtz (d = R): $\frac{8\mu_0 I}{5\sqrt{5}R}$

7.4 Square / polygon / rectangle, at the center

HW 5.8



Setup: n sides, each a segment at distance R with $\theta = \pm\frac{\pi}{n}$

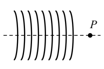
Solve: $B = n \cdot \frac{\mu_0 I}{4\pi R} \cdot 2\sin\frac{\pi}{n}$

$\boxed{\text{n-gon } \frac{n\mu_0 I}{2\pi R}\sin\frac{\pi}{n}\text{; square } \frac{\sqrt{2}\mu_0 I}{\pi R}}$

rectangle $L_1 \times L_2$: $\frac{2\mu_0 I\sqrt{L_1^2+L_2^2}}{\pi L_1 L_2}$

7.5 Finite solenoid, on axis

Disc 5.11



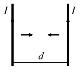
Setup: stack loops: $dI = nI\, dz$

Solve: integrate the loop-axis formula over z

$\boxed{B = \frac{\mu_0 n I}{2}(\cos\theta_2 - \cos\theta_1)\text{; mouth of a semi-infinite one } \frac{\mu_0 n I}{2}}$

7.6 Forces between currents

HW 5.10, Disc 5.4



Setup: $\mathbf{F} = I\int d\mathbf{l} \times \mathbf{B}$, side by side

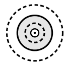
Solve: B of the other wire at each side; sides ⊥ to the wire cancel

$\boxed{\text{parallel wires } f = \frac{\mu_0 I_1 I_2}{2\pi d} \text{ (same direction attract)}}$

square loop near wire: $\frac{\mu_0 I^2 a^2}{2\pi s(s+a)}$ toward; $\mathbf{B} = kz\hat{x}$, loop in yz: $Ika^2\hat{z}$

8 · AMPÈRE & SUPERPOSITION

GENERAL SETUP



$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$

$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$; $\nabla \cdot \mathbf{B} = 0$

USEFUL FACTS

Works only for line, plane, solenoid, toroid symmetry (or sums of them)

Symmetry: $B_s = 0$ (∇·B = 0), $B_z = 0$ (rectangle to ∞) ⇒ $\mathbf{B} = B(s)\hat{\phi}$

Loops: circle (wire), rectangle straddling (sheet), rectangle half inside (solenoid)

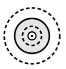
$I_{enc} = \int J(s)\, 2\pi s\, ds$

Inside a uniform wire: $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J} \times \mathbf{s}$ (s from that wire's axis)

Recipe: symmetry sentence → loop → I_{enc} → solve → direction by RHR

SPECIFIC RESULTS (setup → solve → boxed result)

8.1 Solid wire, uniform J



Setup: circle of radius s; $I_{enc} = I\frac{s^2}{a^2}$ (inside), I (outside)

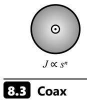
Solve: $B \cdot 2\pi s = \mu_0 I_{enc}$

$\boxed{\text{inside } \frac{\mu_0 I s}{2\pi a^2}\text{; outside } \frac{\mu_0 I}{2\pi s}}$

current only on the surface: 0 inside

8.2 Wire with J ∝ sⁿ; thick tube

HW 5.14



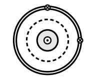
Setup: $I_{enc} = \int_0^s ks'^n\, 2\pi s' ds' = \frac{2\pi k s^{n+2}}{n+2}$

Solve: $B = \frac{\mu_0 I_{enc}}{2\pi s}$

$\boxed{B = \frac{\mu_0 I}{2\pi s}(\frac{s}{a})^{n+2}\text{; } J = ks\text{: } \frac{\mu_0 I s^2}{2\pi a^3}}$

$J = k/s$: $B = \mu_0 k$; tube a..b: $\frac{\mu_0 I}{2\pi s}\frac{s^2-a^2}{b^2-a^2}$

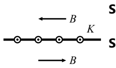
8.3 Coax



Setup: circles in each region; outside, $I_{enc} = I - I = 0$

$\boxed{\text{between } \frac{\mu_0 I}{2\pi s}\text{; outside 0}}$

8.4 Infinite current sheet K



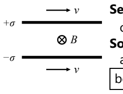
Setup: rectangle straddling the sheet, length ℓ along it; $I_{enc} = K\ell$

Solve: B ∥ sheet, ⊥ K, same size on both sides: $2B\ell = \mu_0 K\ell$

$\boxed{\mathbf{B} = \frac{\mu_0}{2}\mathbf{K} \times \hat{n} \text{ (} \hat{n} \text{ from sheet toward the point)}}$

8.5 Moving capacitor (two opposite sheets)

HW 5.18



Setup: each plate is a sheet $K = \sigma v$, the two in opposite directions

Solve: add the two sheet fields: between they add, outside they cancel

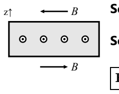
$\boxed{\text{between: } B = \mu_0 \sigma v\text{; outside 0}}$

direction: $\mathbf{K}_{top}\times$(unit vector from top plate into the gap); +σ at y = d, v x̂: $\mathbf{B} = -\mu_0\sigma v\hat{z}$

force/area on top plate: $\frac{\mu_0\sigma^2 v^2}{2}$ (repel) = $\frac{\sigma^2}{2\epsilon_0}$ at v = c

8.6 Thick slab with J x̂, |z|<a

Disc 5.15



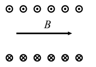
Setup: B(0) = 0 by symmetry; rectangle from the midplane to height z, length ℓ

Solve: $B\ell = \mu_0 Jz\ell$; for |z| > a the enclosed current stops at Ja

$\boxed{\mathbf{B} = -\mu_0 Jz\hat{y} \text{ inside; } \mp\mu_0 Ja\hat{y} \text{ above/below}}$

8.7 Solenoid / toroid

HW 5.17



Setup: solenoid: rectangle, one long side inside (outside B = 0); toroid: circle inside the core

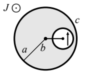
Solve: $BL = \mu_0 nIL$; $B \cdot 2\pi s = \mu_0 NI$

$\boxed{\text{solenoid } \mu_0 nI\text{; toroid } \frac{\mu_0 NI}{2\pi s}}$

Coaxial solenoids, opposite I, turns/length n_1 (inner) and n_2 (outer)

inside both $\mu_0(n_1 - n_2)I$; between $\mu_0 n_2 I$; outside 0

8.8 Cylinder with an off-axis hole



(wire radius a, hole radius c, hole axis at $\mathbf{b} = b\hat{x}$, total current I along +z)

Setup: area ratio: $J = \frac{I}{\pi(a^2-c^2)}$; superpose full cylinder (+J) and hole cylinder (−J)

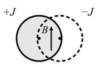
$I_+ = J\pi a^2 = \frac{Ia^2}{a^2-c^2}$, $I_- = -J\pi c^2 = -\frac{Ic^2}{a^2-c^2}$

Solve: inside each: $\frac{\mu_0}{2}\mathbf{J}\times$(vector from its own axis); subtract

$\boxed{\text{in hole: } \mathbf{B} = \frac{\mu_0}{2}\mathbf{J} \times \mathbf{b} = \frac{\mu_0 I b}{2\pi(a^2-c^2)}\hat{y}\text{, uniform}}$

in metal: $\frac{\mu_0}{2}\mathbf{J} \times \mathbf{s} + \frac{\mu_0 I_-}{2\pi s'}\hat{\phi}'$; outside: $\frac{\mu_0 I_+}{2\pi s}\hat{\phi} + \frac{\mu_0 I_-}{2\pi s'}\hat{\phi}'$

8.9 Overlapping wires ±J; slab with a hole



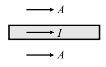
Setup: superpose $\frac{\mu_0}{2}\mathbf{J} \times (\mathbf{s} - \mathbf{c}_+)$ and $-\frac{\mu_0}{2}\mathbf{J} \times (\mathbf{s} - \mathbf{c}_-)$

$\boxed{\text{overlap: } \mathbf{B} = \frac{\mu_0}{2}\mathbf{J} \times (\mathbf{c}_- - \mathbf{c}_+) \text{ uniform}}$

In a hole along x through slab $J\hat{x}$: $-\frac{\mu_0 J}{2}(z\hat{y} + y\hat{z})$

9 · VECTOR POTENTIAL A

GENERAL SETUP



$\mathbf{B} = \nabla \times \mathbf{A}$; Coulomb gauge $\nabla \cdot \mathbf{A} = 0$

$\nabla^2\mathbf{A} = -\mu_0\mathbf{J}$; $\mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}}{|\mathbf{r}-\mathbf{r}'|}d\tau'$ (localized currents only)

$\oint \mathbf{A} \cdot d\mathbf{l} = \Phi_B$

USEFUL FACTS

A points along J

$A_z(s)\hat{z}$: $B_\phi = -\frac{dA_z}{ds}$

$A_\phi(s)\hat{\phi}$: $B_z = \frac{1}{s}\frac{d}{ds}(sA_\phi)$, or $A_\phi \cdot 2\pi s = \Phi$

$A_x(z)\hat{x}$: $B_y = \frac{dA_x}{dz}$

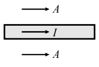
$A_\phi(r,\theta)$: $B_r = \frac{\partial_\theta(\sin\theta A_\phi)}{r\sin\theta}$, $B_\theta = -\frac{\partial_r(rA_\phi)}{r}$

A continuous; $\partial_n\mathbf{A}$ jumps by $-\mu_0\mathbf{K}$; finite on the axis

Recipe: B by Ampère → choose A ∥ J → one curl component = B → integrate → constants from continuity

SPECIFIC RESULTS (setup → solve → boxed result)

9.1 Wire with J = k sⁿ ẑ, A(0) = 0



Setup: Ampère: $B_\phi = \frac{\mu_0 k s^{n+1}}{n+2}$; A = $A_z(s)\hat{z}$

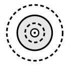
Solve: $-\frac{dA_z}{ds} = B_\phi$; integrate from 0

$\boxed{A_z = -\frac{\mu_0 k s^{n+2}}{(n+2)^2}}$

J = k s²: $B_\phi = \frac{\mu_0 k s^3}{4}$, $\boxed{\mathbf{A} = -\frac{\mu_0 k s^4}{16}\hat{z}}$

check: $\frac{1}{s}(sA_z')' = -\mu_0 J$ ✓

9.2 Uniform wire, everywhere



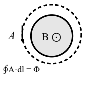
Setup: B inside $\frac{\mu_0 Js}{2}$, outside $\frac{\mu_0 I}{2\pi s}$; choose A(a) = 0

Solve: integrate $-dA_z/ds = B_\phi$ in each region; match at a

$\boxed{\text{inside } \frac{\mu_0 J}{4}(a^2 - s^2)\text{; outside } -\frac{\mu_0 I}{2\pi}\ln\frac{s}{a}}$

coax between: $-\frac{\mu_0 I}{2\pi}\ln\frac{s}{b}$; outside 0

9.3 Solenoid / uniform B



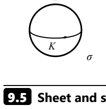
Setup: $\mathbf{A} = A_\phi(s)\hat{\phi}$; Stokes on a circle of radius s

Solve: $A_\phi \cdot 2\pi s = \Phi(s)$

$\boxed{\text{uniform B: } \mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r} = \frac{B_0 s}{2}\hat{\phi}}$

$\boxed{\text{solenoid: } \frac{\mu_0 nIs}{2}\hat{\phi} \text{ in, } \frac{\mu_0 nIR^2}{2s}\hat{\phi} \text{ out}}$

9.4 Spinning shell, dipole, magnetized sphere



Setup: inside uniform B ⇒ $\frac{1}{2}\mathbf{B} \times \mathbf{r}$; outside dipole $\frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{r}}{r^2}$

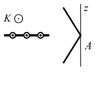
Solve: they match at r = R

$\boxed{\text{shell: in } \frac{\mu_0 R\omega\sigma}{3}r\sin\theta\hat{\phi}\text{, out } \frac{\mu_0 R^4\omega\sigma\sin\theta}{3r^2}\hat{\phi}}$

dipole $\frac{\mu_0 m\sin\theta}{4\pi r^2}\hat{\phi}$; M-sphere inside $\frac{\mu_0 M}{3}r\sin\theta\hat{\phi}$

9.5 Sheet and slab

HW 5.29



Setup: $\mathbf{A} = A_x(z)\hat{x}$, $B_y = \frac{dA_x}{dz}$

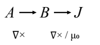
Solve: integrate the sheet/slab B; A continuous

$\boxed{\text{sheet } K\hat{x}\text{: } -\frac{\mu_0 K}{2}|z|\hat{x}}$

slab $J\hat{x}$: in $-\frac{\mu_0 Jz^2}{2}$; out $-\mu_0 Ja|z| + \frac{\mu_0 Ja^2}{2}$

9.6 Given A: find B, then J

HW 5.26



Setup: curl in the matching coordinates (one component)

Solve: $\mathbf{B} = \nabla \times \mathbf{A}$, then $\mathbf{J} = \frac{1}{\mu_0}\nabla \times \mathbf{B}$

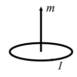
$\boxed{A = k\hat{\phi}\text{: } B = \frac{k}{s}\hat{z}\text{, } J = \frac{k}{\mu_0 s^2}\hat{\phi}}$

$A = ks^3\hat{z}$: $B = -3ks^2\hat{\phi}$, $J = -\frac{9ks}{\mu_0}\hat{z}$

Finite wire, ends at z_1, z_2 (measured from P): $A_z = \frac{\mu_0 I}{4\pi}\ln\frac{z_2+\sqrt{z_2^2+s^2}}{z_1+\sqrt{z_1^2+s^2}}$

10 · MAGNETIC DIPOLES & SPINNING CHARGE

GENERAL SETUP



$\mathbf{m} = I\mathbf{a}$ (RHR: fingers with I, thumb = m)

$\mathbf{B} = \frac{\mu_0 m}{4\pi r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta})$; $\mathbf{A} = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{r}}{r^2}$

USEFUL FACTS

$\mathbf{N} = \mathbf{m} \times \mathbf{B}$; $U = -\mathbf{m} \cdot \mathbf{B}$; $\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$

No magnetic monopole term

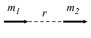
Any spinner: $m = \frac{Q}{2M}L$: ring ½, disk ¼, shell ⅓, ball ⅕ (× QωR²)

Average B inside a sphere: $\frac{\mu_0}{4\pi}\frac{2\mathbf{m}}{R^3}$

SPECIFIC RESULTS (setup → solve → boxed result)

10.1 Two dipoles

HW 6.3



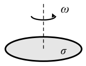
Setup: B of m_1 on its axis: $\frac{\mu_0}{4\pi}\frac{2m_1}{z^3}$

Solve: $F = m_2\frac{dB}{dz}$

$\boxed{\text{coaxial: } F = \frac{3\mu_0 m_1 m_2}{2\pi r^4} \text{ (attract)}}$

10.2 Spinning disk (σ, ω)

Disc 5.39



Setup: rings of radius s, width ds: $dI = \sigma\omega s\, ds$

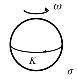
Solve: $m = \int \pi s^2 dI$; center $B = \int \frac{\mu_0 dI}{2s}$

$\boxed{m = \frac{\pi}{4}\sigma\omega R^4\text{; center } B = \frac{\mu_0\sigma\omega R}{2}}$

axis: $\frac{\mu_0\sigma\omega}{2}[\frac{R^2+2z^2}{\sqrt{R^2+z^2}} - 2z]$

10.3 Spinning shell (σ, ω)

Ex 5.11



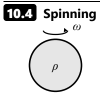
Setup: $K = \sigma\omega R\sin\theta$; outside is a pure dipole

Solve: $m = \frac{Q}{2M}L$; inside from the BC at the equator: $\mathbf{B}_{out} - \mathbf{B}_{in} = \mu_0\mathbf{K} \times \hat{r}$

$\boxed{\text{inside } \mathbf{B} = \frac{2}{3}\mu_0\sigma\omega R\hat{z} \text{ (uniform); } m = \frac{4}{3}\pi\sigma\omega R^4}$

10.4 Spinning solid ball (ρ, ω)

Disc 5.63



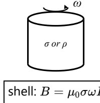
Setup: shells of radius r: each a spinning shell with $\sigma = \rho\, dr$

Solve: $m = \int dm_{shell}$; center: add the uniform inside fields

$\boxed{m = \frac{1}{5}Q\omega R^2\text{; center } \frac{\mu_0 Q\omega}{4\pi R}\text{; average } \frac{\mu_0 Q\omega}{10\pi R}}$

$\mathbf{B}_{in} = \mu_0\rho\omega[(\frac{R^2}{3} - \frac{r^2}{5})\cos\theta\hat{r} - (\frac{R^2}{3} - \frac{2r^2}{5})\sin\theta\hat{\theta}]$

10.5 Spinning cylinders

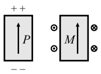


Setup: shell: $K = \sigma\omega R$ = solenoid; solid: nested shells

$\boxed{\text{shell: } B = \mu_0\sigma\omega R \text{ inside, } m = \pi\sigma\omega R^3 L\text{; solid: } B(s) = \frac{\mu_0\rho\omega}{2}(R^2 - s^2)}$

11 · B, H, M: MAGNETIC MATTER

GENERAL SETUP



$\mathbf{M}$ = magnetic moment per volume (A/m)

Bound current: $\mathbf{J}_b = \nabla \times \mathbf{M}$, $\mathbf{K}_b = \mathbf{M} \times \hat{n}$

$\mathbf{H} \equiv \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$: $\nabla \times \mathbf{H} = \mathbf{J}_f$, $\oint \mathbf{H} \cdot d\mathbf{l} = I_{f,enc}$

$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ (NOT zero in general)

USEFUL FACTS

Linear: $\mathbf{M} = \chi_m\mathbf{H}$ and $\mathbf{B} = \mu\mathbf{H}$

$\mu = \mu_0(1 + \chi_m) = \mu_0\mu_r$; $\mu_r = 1 + \chi_m$

Homogeneous: $\mathbf{J}_b = \chi_m\mathbf{J}_f$

Para: $\chi_m > 0$ (~10⁻⁵), pulled in; dia: $\chi_m < 0$, pushed out; ferro: keeps M

No free current + symmetry ⇒ H = 0 ⇒ $\mathbf{B} = \mu_0\mathbf{M}$

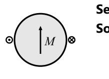
Free current: H from Ampère, then $\mathbf{B} = \mu\mathbf{H}$ (or $\mu_0(\mathbf{H} + \mathbf{M})$)

Shape: $\mathbf{B}_{in} = \mu_0(1 - N)\mathbf{M}$, $\mathbf{H}_{in} = -N\mathbf{M}$

P charges the caps; M puts current on the sides

SPECIFIC RESULTS (setup → solve → boxed result)

11.1 Uniformly magnetized sphere



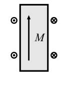
Setup: $\mathbf{J}_b = 0$; $\mathbf{K}_b = \mathbf{M} \times \hat{r} = M\sin\theta\hat{\phi}$

Solve: same as a spinning shell with $\sigma\omega R \to M$

$\boxed{\mathbf{B}_{in} = \frac{2}{3}\mu_0\mathbf{M}\text{, } \mathbf{H}_{in} = -\frac{\mathbf{M}}{3}\text{; outside dipole } \frac{4}{3}\pi R^3\mathbf{M}}$

uniform-M shell: B = 0 in the cavity; in an external $\mathbf{B}_0$: add it

11.2 Magnetized rod / disk



Setup: uniform M along the axis: $\mathbf{K}_b = M\hat{\phi}$ on the side, nothing on the caps

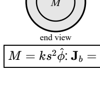
Solve: long rod = solenoid with nI → M

$\boxed{\text{long rod: } B = \mu_0 M\text{; thin disk: B ≈ 0, H ≈ −M}}$

finite rod on its axis: $\frac{\mu_0 M}{2}(\cos\theta_2 - \cos\theta_1)$

11.3 Cylinder with M along φ̂

HW 6.8



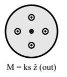
Setup: $\mathbf{J}_b = \frac{1}{s}\frac{d}{ds}(sM)\hat{z}$; $\mathbf{K}_b = \mathbf{M} \times \hat{s}$

Solve: no free current + symmetry ⇒ H = 0 ⇒ $\mathbf{B} = \mu_0\mathbf{M}$

$\boxed{M = ks^2\hat{\phi}\text{: } \mathbf{J}_b = 3ks\hat{z}\text{, } \mathbf{K}_b = -kR^2\hat{z}\text{, } \mathbf{B} = \mu_0\mathbf{M} \text{ in, 0 out}}$

11.4 Cylinder with M = k s ẑ

HW 6.12



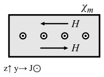
Setup: $\mathbf{J}_b = -\frac{dM_z}{ds}\hat{\phi}$; $\mathbf{K}_b = \mathbf{M} \times \hat{s}$

Solve: rectangle half inside: $\oint \mathbf{H} \cdot d\mathbf{l} = 0$ ⇒ H = 0 ⇒ $\mathbf{B} = \mu_0\mathbf{M}$

$\boxed{\mathbf{J}_b = -k\hat{\phi}\text{, } \mathbf{K}_b = kR\hat{\phi}\text{, } \mathbf{B} = \mu_0 ks\hat{z} \text{ in, 0 out}}$

plus a free wire I on the axis: $H = \frac{I}{2\pi s}$, $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$

11.5 Linear slab carrying free J₀ x̂, |z|<d/2



Setup: $\mathbf{H} = H(z)\hat{y}$, odd in z; rectangle from −z to z, length ℓ (counterclockwise from +x)

Solve: $-2H\ell = J_0(2z)\ell$ ⇒ H; then $\mathbf{M} = \chi_m\mathbf{H}$, $\mathbf{B} = \mu\mathbf{H}$, $\mathbf{J}_b = \nabla \times \mathbf{M}$, $\mathbf{K}_b = \mathbf{M} \times \hat{n}$

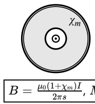
$\boxed{\mathbf{H} = -J_0 z\hat{y}\text{; } \mathbf{M} = -\chi_m J_0 z\hat{y}\text{; } \mathbf{B} = -\mu_0(1 + \chi_m)J_0 z\hat{y}}$

$\boxed{\mathbf{J}_b = \chi_m J_0\hat{x}\text{; } \mathbf{K}_b = -\frac{\chi_m J_0 d}{2}\hat{x} \text{ on both faces}}$

outside: $\mathbf{H} = \mp\frac{J_0 d}{2}\hat{y}$, $\mathbf{B} = \mu_0\mathbf{H}$; net bound current 0 ✓

11.6 Coax / toroid / solenoid with χₘ

HW 6.16



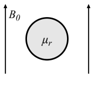
Setup: Ampère for H with free current only

Solve: $H = \frac{I}{2\pi s} \to B = \mu H \to M = \chi_m H \to \mathbf{K}_b = \mathbf{M} \times \hat{n}$

$\boxed{B = \frac{\mu_0(1+\chi_m)I}{2\pi s}\text{, } M = \frac{\chi_m I}{2\pi s}\text{; inner surface carries } I_b = \chi_m I}$

core in a solenoid/toroid: B × (1 + χ_m); half-cylinders $\mu_1|\mu_2$: $B = \frac{\mu_1\mu_2}{\mu_1+\mu_2}\frac{I}{\pi s}$

11.7 Linear sphere / slab in uniform B₀



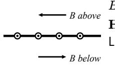
Setup: same as the dielectric sphere with $\epsilon \to \mu$, $\mathbf{E} \to \mathbf{H}$

$\boxed{\text{sphere } \mathbf{B}_{in} = \frac{3\mu_r}{\mu_r+2}\mathbf{B}_0}$

slab ⊥ B_0: B same, H ÷ μ_r; slab ∥ B_0: H same, B × μ_r

12 · MAGNETIC BOUNDARY CONDITIONS & E ↔ B

GENERAL SETUP



$B^\perp$ continuous; $\mathbf{B}_{above} - \mathbf{B}_{below} = \mu_0\mathbf{K} \times \hat{n}$

$\mathbf{H}^\parallel$ jumps by $\mathbf{K}_f \times \hat{n}$

Linear media: $\mu_1 H_1^\perp = \mu_2 H_2^\perp$; with no free K, $\mathbf{H}^\parallel$ is continuous too

A continuous; $\partial_n\mathbf{A}$ jumps by $-\mu_0\mathbf{K}$

USEFUL FACTS

ρ ↔ J; ϵ_0 ↔ $1/\mu_0$; V ↔ A (per component)

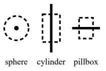
Gauss ↔ Ampère; $\frac{p}{4\pi\epsilon_0}$ ↔ $\frac{\mu_0 m}{4\pi}$; D ↔ H; P ↔ M

Sphere: $-\frac{\mathbf{P}}{3\epsilon_0}$ ↔ $+\frac{2}{3}\mu_0\mathbf{M}$; long rod: ≈ 0 ↔ $\mu_0 M$

Fill all space: E ÷ ϵ_r ↔ B × μ_r

13 · EXAM I BASICS: COULOMB, GAUSS, CONDUCTORS

GENERAL SETUP



$\mathbf{E} = k\int \frac{dq\,(\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|^3}$, $V = k\int \frac{dq}{|\mathbf{r}-\mathbf{r}'|}$, $dq = \lambda dl, \sigma da, \rho d\tau$

$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$; $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$; $\nabla^2 V = -\frac{\rho}{\epsilon_0}$

$V(b) - V(a) = -\int_a^b \mathbf{E} \cdot d\mathbf{l}$; $W = q\Delta V$

USEFUL FACTS

Conductor: E = 0 inside, V constant, charge on the surface, $E = \sigma/\epsilon_0$ just outside

Charge q in a cavity ⇒ −q on the cavity wall

$E^\perp$ jumps by σ/ϵ_0; $E^\parallel$ and V continuous

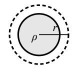
Given E: $\rho = \epsilon_0\nabla \cdot \mathbf{E}$; $Q_{enc}(r) = 4\pi\epsilon_0 r^2 E_r$; ≠ 0 as r → 0 ⇒ point charge

$\nabla \cdot \frac{\hat{r}}{r^2} = 4\pi\delta^3(\mathbf{r})$; $\nabla \cdot \frac{\hat{s}}{s} = 2\pi\delta^2$

Exam I classics: plane at V_0 + charge → 4.1; grounded sphere in E_0 → 3.2; slab + sphere overlap → 13.1 + 13.3 (add fields); energy integral → 2.8; div/curl/V → 6

SPECIFIC RESULTS (setup → solve → boxed result)

13.1 Point charge / ball



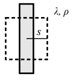
Setup: Gaussian sphere radius r

Solve: $E \cdot 4\pi r^2 = Q_{enc}/\epsilon_0$

$\boxed{\text{outside } \frac{kQ}{r^2}\text{; inside uniform ball } \frac{\rho r}{3\epsilon_0}\text{; } V_{in} = \frac{kQ}{2R}(3 - \frac{r^2}{R^2})}$

$\rho = kr^n$: $E = \frac{kr^{n+1}}{(n+3)\epsilon_0}$

13.2 Line / cylinder



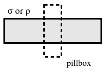
Setup: Gaussian cylinder radius s, length L

Solve: $E \cdot 2\pi sL = \lambda_{enc}L/\epsilon_0$

$\boxed{\text{line } \frac{\lambda}{2\pi\epsilon_0 s}\text{; inside uniform cylinder } \frac{\rho s}{2\epsilon_0}}$

$\rho = ks^n$: $\frac{ks^{n+1}}{(n+2)\epsilon_0}$; $V = -\frac{\lambda}{2\pi\epsilon_0}\ln\frac{s}{s_0}$

13.3 Sheet / slab

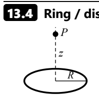


Setup: pillbox straddling the sheet (or from the midplane of a slab)

Solve: $2EA = \sigma A/\epsilon_0$

$\boxed{\text{sheet } \frac{\sigma}{2\epsilon_0} \text{ each side; conductor surface } \frac{\sigma}{\epsilon_0}\text{; slab |z|<a: } \frac{\rho z}{\epsilon_0} \text{ in, } \frac{\rho a}{\epsilon_0} \text{ out}}$

13.4 Ring / disk / segment, on axis



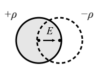
Setup: Coulomb integral; by symmetry only the axial component survives

Solve: $E_z = k\int \frac{z\,dq}{(z^2+R^2)^{3/2}}$; disk = rings of radius s

$\boxed{\text{ring } \frac{kQz}{(z^2+R^2)^{3/2}}\text{; disk } \frac{\sigma}{2\epsilon_0}(1 - \frac{z}{\sqrt{z^2+R^2}})}$

segment: $E_\perp = \frac{k\lambda}{s}(\sin\theta_2 - \sin\theta_1)$

13.5 Overlaps and cavities



Setup: hole = whole shape + negative copy, each field from its own center

$\boxed{\text{cavity offset } \mathbf{d} \text{ in a ball ρ: } \mathbf{E} = \frac{\rho\mathbf{d}}{3\epsilon_0} \text{ uniform}}$